

An Efficient Spectral Clustering Algorithm Based on Granular-Ball

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Abstract—In order to solve the problem that the traditional spectral clustering algorithm is time-consuming and resource consuming when applied to large-scale data, resulting in poor clustering effect or even unable to cluster, this paper proposes a spectral clustering algorithm based on granular-ball (GBSC). The algorithm changes the construction method of the similarity matrix. Based on granular-ball, the size of the similarity matrix is greatly reduced, and the construction of the similarity matrix is more reasonable. Experimental results show that the proposed algorithm achieves better speedup ratio, less memory consumption and stronger anti noise performance while achieving similar clustering results to the traditional spectral clustering algorithm. Suppose the number of granular-balls is m , n is the number of points in the dataset, and $m \ll n$, the time complexity of GBSC is $O(m^3)$. It is proved that GBSC has good adaptability to large-scale datasets. All codes have been released at <https://github.com/xjnine/GBSC>.

Index Terms—Clustering, granular computing, granular-ball, spectral clustering.

I. INTRODUCTION

CLUSTERING is the unsupervised classification of patterns (observations, data items, or feature vectors) into groups (clusters) [1]. Clustering has been widely used in research fields such as data mining [2], pattern recognition [3], image analysis [4], and information retrieval [5]. After decades of research, a large number of clustering algorithms have been proposed, such as partition clustering [6], hierarchical clustering [7], density clustering [8], grid clustering [9], model clustering [10], and graph clustering [11]. As a representative algorithm in graph clustering algorithms, spectral clustering has also been widely studied. In the beginning, people's research on spectral clustering mainly focused on graph partitioning criteria, and more attention was paid to how to divide clusters more accurately,

and developed the minimum cut set criterion [12], the normative cut set criterion [13], the proportional cut set criterion [14], the average cut set criterion [15], the maximum and minimum cut set criterion [16], the multi-path normative cut set criterion [17] and the division method of spectral clustering algorithm has been very perfect. However, as the scale of data increases, the shortcomings of spectral clustering begin to become prominent. The eigen decomposition calculation complexity of spectral clustering is high, and its time complexity is as high as $O(n^3)$.

Therefore, how to improve the performance of spectral clustering has been widely studied. Zhang [18] et al. gives consideration to the quality of intra view division and the collaboration between views, carries out consistency constraints on diversified large-scale data sets through multi representative point strategy and multi view representativeness, and realizes spectral clustering of data through maximization of weighting coefficient and Lagrange multiplier. Compared with traditional spectral clustering, the improved algorithm has certain advantages in clustering time and storage optimization. Jiang [19] et al. selects nonnegative matrix decomposition to process multi view data to obtain the potential feature matrix of each view, then introduces orthogonal constraints to optimize local features, adjusts the clustering contribution of missing data through data weighting, and finally alleviates the memory requirements of large-scale data through block processing. Fowlkes [20] et al. uses Nyström approximate sampling to reduce the complexity of similarity calculation of large-scale data, which is conducive to the improvement of clustering efficiency. Through the adaptive optimization of the sampling process, the clustering effect of spectral clustering on large-scale data sets is further improved. Huang [21] et al. proposes a spectral clustering method based on K-means approximation. First, K-means clustering is performed on the data, and then spectral clustering is performed on the obtained clustering centers (key nodes). Other data points are assigned to the corresponding clusters following their corresponding clustering centers (key nodes). Cai [22] et al. re-encodes the randomly selected key nodes and the original data, and performs singular value decomposition (SVD) on the encoded new data.

Most of the existing clustering methods are based on a single granularity of information, such as the distance and density of each data. Therefore, the clustering algorithm is greatly affected by outliers, and the process of calculate the Laplace matrix L is slow. Although some research achievements have been made in spectral clustering on large-scale data sets. Most of the selected key nodes are obtained by random selection or K-means

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clustering, which can not guarantee that the selected key nodes can well represent the original data, thus affecting the subsequent clustering results.

In order to solve the above problems, this paper proposes an improved spectral clustering algorithm based on granular-ball, which is a multi-granularity structure and can adaptively characterize unlabeled data [23]. We use granular-ball to represent unsupervised data, and then process the data efficiently with appropriate granularity. The core idea of granular-ball is that data with similar distribution form a granular-ball, and adjacent granular-balls form a cluster. We use granular-ball to represent the data in a coarse-grained way. We propose a clustering algorithm that combines multi-granularity Granular-Ball and Spectral Clustering.

The main contributions of this paper are as follows:

- 1) Self-adaption: The generation of granular-ball is completely parameter-free. Therefore, the algorithm does not require manual parameters.
- 2) Efficiency: Based on granular-ball, the size of the similarity matrix is greatly reduced and the construction of the similarity matrix is more reasonable. Suppose the number of granular-balls is m , n is the number of points in the dataset, and $m \ll n$, the time complexity of GBSC is $O(m^3)$.
- 3) Robustness: Since each granular-ball covers many points and only contains two data, namely center and radius, a small number of noise points can be smoothed, so that the granular-ball will not be affected.

II. RELATED WORK

A. Large Scale Spectral Clustering

The spectral clustering method transforms the data clustering problem into a graph partition problem by using graph theory. By decomposing the eigenvalues of the graph Laplace matrix, the vector representation of the original data in the transformed low dimensional space is obtained. Finally, the k-means algorithm is run on these low dimensional feature vectors to get the final clustering results.

Among them, the time complexity defects of spectral clustering are mainly reflected in three aspects: the construction of similarity matrix $O(n^2 d)$, the eigenvalue decomposition of Laplace matrix $O(n^3)$ and the final K-means clustering step $O(nkt)$ (t is the number of iterations). Its spatial complexity defect is mainly reflected in the $O(n^2)$ required to store the similarity matrix and Laplace matrix. With the increase of data size n , the computational complexity of spectral clustering is too high, which makes the spectral clustering method unable to play a better role in dealing with large-scale data sets.

In view of the above limitations on the application of spectral clustering to large-scale data sets, the existing spectral clustering methods can be divided into two types: feature map-based methods and sample reduction-based methods.

Hansen [24] et al. propose a large-scale spectral clustering algorithm based on the feature mapping method. This algorithm is proposed based on the fact that the feature vector is not suitable

for the case of being interested in local features of data. It provides a method to construct Laplacian semi supervised feature vector to solve this problem. The algorithm uses locally biased feature vectors to perform locally biased machine learning, which overcomes the problem of applying spectral clustering to large-scale data. Wu [25] et al. proposed SCRB algorithm, it is the first time to use the random bin feature (RB) to generate the inner product of a large-scale sparse feature matrix, use the inner product to implicitly approximate the graph similarity matrix, and introduce the singular value decomposition solver to calculate the feature vector of the large-scale matrix, so as to accelerate the construction and decomposition of the similarity graph. Rahman [26] et al. Combined histogram based features with distribution based density features to develop a flexible feature mapping technology, which contains prior knowledge of data and provides flexible representation for it, thus improving the recognition ability of the classifier. Vazquez Martin [27] et al. proposed a key frame detector. The algorithm also establishes auxiliary mapping based on local characteristics, and then uses spectral clustering method to obtain graph partition. This method does not need to estimate the motion of the camera, and the similarity measure defined in the detector can be used for any type of feature. Yang [28] et al. proposed a local spectral clustering mapping algorithm based on sparse self coding. By preprocessing the data, the algorithm uses sparse self coding to extract deep-seated features that can reflect the nature of the original data, and replaces the original data. Each data is linearly reconstructed by its neighborhood, and the reconstruction weight is used to replace the Gaussian kernel function to establish the adjacency matrix.

Another kind of large-scale spectral clustering algorithm is to reduce the size of sample data set by selecting representative objects. This method reduces the size of available data sets by selecting representative objects in the data set and using the representative objects to form a small-scale data set. Then, the existing spectral clustering method is used to partition the small-scale data set composed of these representative objects. Finally, the class to which the data in the original dataset belongs is allocated according to the class to which the representative object belongs. The typical method is to use Nystrom method to reduce the computational cost of solving eigenvectors. The main idea is that given a data matrix, the Nystrom method is used to randomly sample the data, calculate the eigenvectors of the sub matrix, and use the calculated eigenvectors to estimate the eigenvectors of the original large-scale matrix [29]. For example, Charles [20] et al. Applied the Nystrom method to the grouping problem of permutation and combination. Based on the fact that the pixels occupied by the target in the real scene are far less than all pixels, it is just consistent with the random sampling data of Nystrom method, and successfully solves the grouping problem of large-scale data. In addition, He [30] et al. proposed an algorithm. The main idea is to use random Fourier features to explicitly represent data in kernel space. It solves the problem of insufficient memory by constructing a kernel matrix which is far smaller than the data scale. At the same time, the running speed of the algorithm is significantly faster.

In addition, there are some large-scale spectral clustering algorithms that are not based on the above two representative

methods, such as the algorithm proposed by Huang [21] et al. It adopts a mixed representation selection strategy, and designs a k-nearest representation fast approximation method to construct a sparse affinity submatrix, then interprets the sparse submatrix as a bipartite graph, and finally uses transfer cutting to effectively partition the graph to obtain clustering results. Yang [31] et al. applied the concept of hypergraph to the improvement of spectral clustering algorithm, thus proposed an algorithm, which extended hypergraph to a general format to capture complex high-order relationships, and proposed the concept of “feature technique” to reduce the computational complexity and accelerate the solving process of feature problems. This algorithm overcomes the problem of insufficient memory on large-scale data by storing only hypergraph instance matrix, but not Laplace matrix.

It can be seen that sample extraction is a main strategy for optimizing the similarity matrix of spectral clustering methods. Although the operation of reference sampling using the distance of data points is intuitive and simple, the artificial selection of threshold and the mapping of feature space will lead to the difference between representative samples and actual samples. In addition, using the method of binding representative points to sparse the graph will damage the density of the data point set and reduce the clustering accuracy. Increasing the number of representative points according to the size of the data set will result in that the larger the amount of data, the more representative points, and the lower the accuracy of the algorithm.

The generation process of granular-ball is a process of adaptive characterization of unlabeled data, which can eliminate the interference of noise and reduce the scale of data. This paper first optimizes the generation process of granular-ball, and then uses granular-ball to improve the construction of similarity matrix and improve the efficiency of spectral clustering algorithm.

B. The Basic Process of Spectral Clustering

The spectral clustering algorithm adopts the idea of undirected graph. The data in the data set is regarded as the nodes in the undirected graph, and the similarity between the data is expressed as the weight between the nodes. Then the undirected graph is divided into different subgraphs according to the weight rules. The clustering analysis of the data is realized by optimizing the high weight of the edges in the subgraph and the low weight of the edges between the subgraphs. The solution of optimal division of spectral clustering usually depends on Laplace matrix, and its process is as follows:

- 1) Input the initial class number k and the similarity matrix $s \in R^{(n \times m)}$, then establish the similarity graph of the data set, and calculate the edge weight matrix W ;
- 2) Calculate the Laplace matrix L of the graph, and give the matrix composed of the first k eigenvectors: $V = [v_1, v_2, \dots, v_k] \in R^{(n \times k)}$; Normalize each row of V so that its norm value is 1 to obtain the matrix $U = [u_{ij}]$, where,

$$u_{ij} = \frac{v_{ij}}{(\sum_k v_{ik}^2)^{1/2}} \quad (1)$$

- 3) Set $y_i \in R^k$ corresponds to column i of matrix U , $i = 1, 2, \dots, n$, then $y_i \in R^k$ for clustering, clustering result $C_1, C_2, \dots, C_k$.

C. Granular-Ball Computing

Given a data set $D = p_i (i = 1, 2, \dots, n)$, where n is the number of samples on D . Granular balls $GB_1, GB_2, \dots, GB_m$ are used to cover and represent the data set D . Suppose the number of samples in the j^{th} granular-ball GB_j is expressed as $|GB_j|$, then its coverage degree can be expressed as $\sum_{j=1}^m (|GB_j|)/n$. The basic model of granular-ball coverage can be expressed as

$$\begin{aligned} \min \quad & \lambda_1 * n / \sum_{j=1}^m (|GB_j|) / n + \lambda_2 * m, \\ \text{s.t.} \quad & \text{quality}(GB_j) \geq T, \end{aligned} \quad (2)$$

where λ_1 and λ_2 are the corresponding weight coefficients, and m the number of granular balls. When other factors remain unchanged, the higher the coverage, the less the sample information is lost, and the more the number of granular-balls, the the characterization is more accurate. Therefore, the minimum number of granular-balls should be considered to obtain the maximum coverage degree when generating granular-balls. By adjusting the parameters λ_1 and λ_2 , the optimal granular-ball generation results can be obtained to minimize the value of the whole equation. In most cases, the two items in the objective function do not affect each other and do not need trade off, so λ_1 and λ_2 are set to 1 by default. Granular-ball computing can fit arbitrarily distributed data [34], [37], [39].

III. GRANULAR-BALL SPECTRAL CLUSTERING

A. The Process of Generating Granular-Balls

The granular-ball method proposed by Xia [23] et al. treat the entire data as the most coarse-grained, and use granular-ball to split and refine from coarse-grained to fine-grained to achieve a scalable and robust computing process. This approach is based on a basic assumption that the data with similar distribution form a granular-ball and the adjacent granular-balls form a cluster. In this paper, the splitting standard is changed from that the quality of the child ball is better than that of the parent ball to that of the weighted child ball. This improvement can obtain a more reasonable set of granular-balls in noisy datasets.

Definition 1. Given a dataset $D \in R^d$, there are two quantities: Granular-Ball(GB_j) and Distribution Measure(DM_j). For each GB_j , the c_j is the center of GB_j and r_j is the radius of GB. The definitions of c_j and r_j are as follows:

$$c_j = \frac{1}{n_j} \sum_{i=1}^{n_j} p_i. \quad (3)$$

$$r_j = \max(|p_i - c_j|), \quad (4)$$

where $||\cdot||$ denotes the 2-norm and n_j represents the number of data located in GB_j .

Definition 2. DM_j is measured by computing the ration of the number data point n_j and the sum radius s_j in GB_j . Where

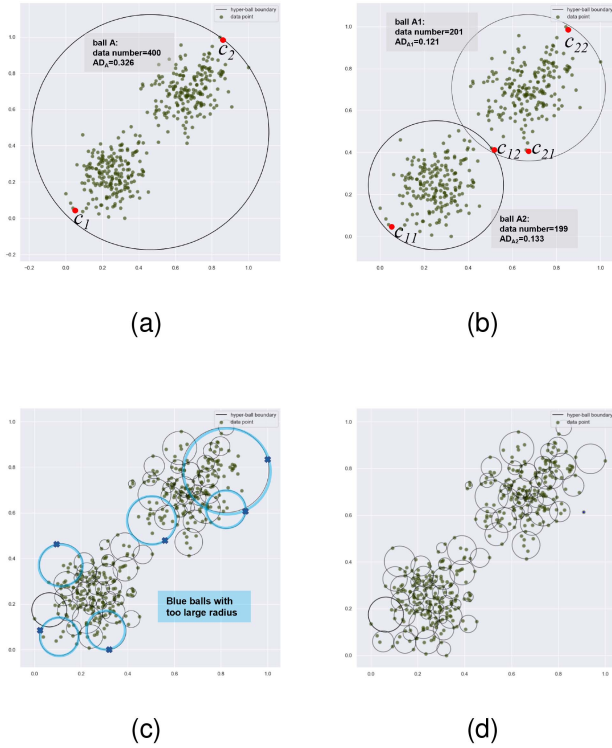


Fig. 1. The process of generating granular-balls on a simplified example. (a) Raw data points in granular-ball A. (b) Child granular-ball of A. (c) Blue balls with radii that are too large. (d) The final granular-ball sets after removing granular-ball with a radius that is too large.

$$s_j = \sum_{i=1}^{n_j} \|p_i - c_j\|, \quad DM_j = \frac{s_j}{n_j}. \quad (5)$$

The granular-ball was divided according to DM . As shown in Fig. 1: First, we treat the whole dataset as a granular-ball A ; Then, we choose the two farthest points C_1 and C_2 to split the ball A into two sub-balls A_1 and A_2 ; third, We calculate the DM_A , DM_{A_1} and DM_{A_2} values for A , A_1 and A_2 ; At last, We judge whether to split the ball by comparing DM_A and DM_{A_1} with DM_{A_2} . In [23], both DM_{A_1} and DM_{A_2} need to be smaller than DM_A for GB_A to split. However, when there is a lot of noise, this splitting rule will cause a large number of granular-balls to fail to split.

Definition 3. (Weighted DM value). Therefore, we use a weighted DM value for comparison in this paper, which can better adapt to noisy situations. DM_{weight} is defined as follows:

$$DM_{weight} = \frac{n_{A1}}{n_A} DM_{A1} + \frac{n_{A2}}{n_A} DM_{A2}. \quad (6)$$

where n_A , n_{A_1} and n_{A_2} represent the number of data in the corresponding granular-ball, respectively. If DM_{weight} is smaller than DM_A , the granular-ball A splits. In Fig. 1(c), the final splitting result is illustrated. However, in Fig. 1(c), some granular-balls with radii that are too large may still be affected by some boundary or noise points and need to be split; if $r_j > 2 \times \max(\text{mean}(r), \text{median}(r))$, GB_j needs to be split. $\text{mean}(r)$ and $\text{median}(r)$ represent the mean and median of

all granular-ball radii, respectively. After granular-balls with radii that are too large are removed, the splitting process is completed, and the result is shown in Fig. 1(d). Based on the above descriptions, the generation of granular-balls algorithm is shown in Algorithm 1.

B. Construction of Similarity Matrix Based on Granular Ball

When dividing undirected graph, it is necessary to measure the distance between data through Gaussian kernel function, and then form the similarity connection graph of data. The similarity graph corresponds to a similarity matrix. Generally, data mining needs to deal with diversified data, so the similarity graph usually adopts full connection graph. Given a sample point $p_i (i = 1, 2, \dots, n)$, the number of data set is n , so the edge weight of the undirected graph calculated from the data distance value is:

$$e_{ij} = e(p_i, p_j) = \sum_{i=1, j=1}^n \exp \frac{-\|p_i - p_j\|^2}{2\sigma^2} \quad (7)$$

The similarity matrix is as follows:

$$W = \begin{bmatrix} e_{11} & e_{12} & e_{13} & e_{14} & \cdots & e_{1n} \\ e_{21} & e_{22} & e_{23} & e_{24} & \cdots & e_{2n} \\ e_{31} & e_{32} & e_{33} & e_{34} & \cdots & e_{3n} \\ e_{41} & e_{42} & e_{43} & e_{44} & \cdots & e_{4n} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ e_{n1} & e_{n2} & e_{n3} & e_{n4} & \cdots & e_{nn} \end{bmatrix} \quad (8)$$

The similarity matrix construction process based on granular-balls is shown in Fig. 2. The similarity matrix W is constructed by using granular-ball theory. The distance of two granular-balls is defined as the distance between the centers of the two granular-balls minus the radii of the two granular-balls. If the two balls overlap, the distance is set 0. The granular-balls participating in the construction of the similarity matrix do not include all granular-balls belonging to the outlier set. The distance $dis(GB_{j1}, GB_{j2})$ and outlier set are defined as follows:

$$dis(GB_{j1}, GB_{j2}) = dis(c_{j1}, c_{j2}) - (r_{j1} + r_{j2}). \quad (9)$$

$$outlier = \{GB_j | n_j \leq 2\}. \quad (10)$$

The number of granular-balls generated from the data set is m , which is represented by $b_j (j = 1, 2, \dots, m)$, so the edge weight of the new calculated undirected graph is:

$$b_{ij} = e(b_i, b_j) = \sum_{i=1, j=1}^m \exp \frac{-\|b_i - b_j\|^2}{2\sigma^2} \quad (11)$$

In this way, the granular-ball similarity matrix can be obtained.

$$W = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ b_{k1} & b_{k2} & b_{k3} & b_{mm} \end{bmatrix} \quad (12)$$

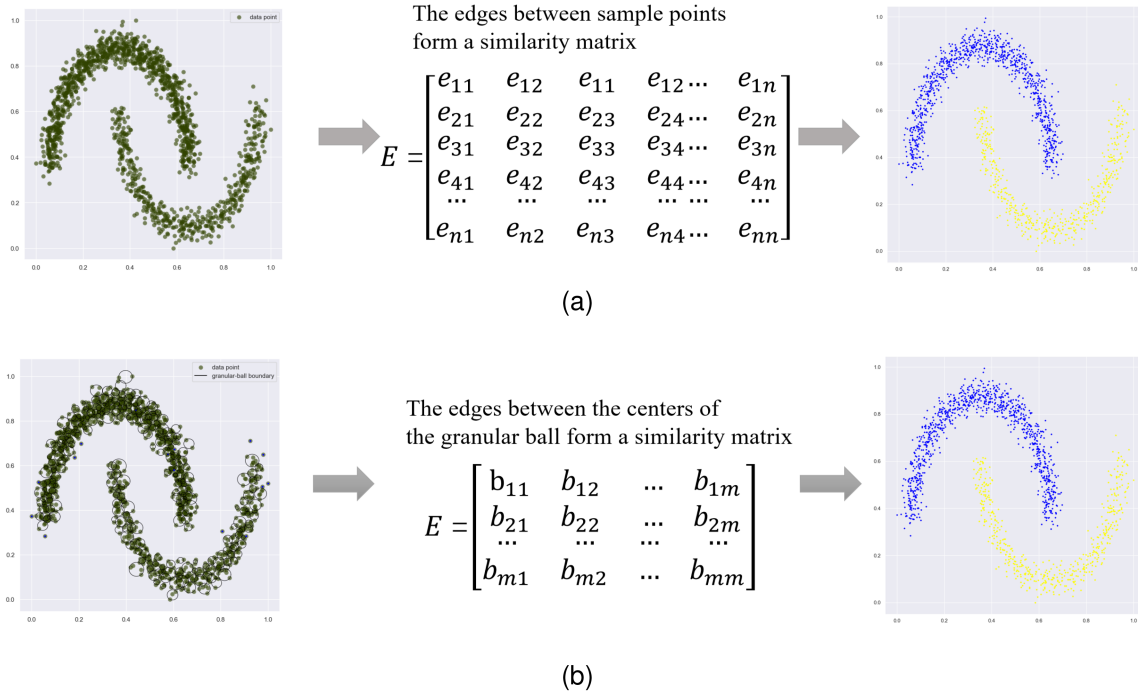


Fig. 2. Original spectral clustering and granular-ball based spectral clustering.

C. Improved Spectral Clustering Algorithm Based on Granular Ball

The GBSC algorithm mainly includes the following steps: first, processing raw data into multi granularity granular-ball set; second, constructing spectral clustering algorithm based on granular-ball set similarity matrix. According to the above description, the GBSC algorithm is shown in algorithm 1 and algorithm 2. GBSC mainly includes two stages: generating granular-ball and spectral clustering based on granular-ball. Suppose the number of granular-balls is m , n is the number of points in the dataset, and l is the number of edges. The time complexity of generating granular-ball is $O(n \log n)$. The construction time complexity of similarity matrix based on granular-ball spectral clustering is $O(m^2)$, and the eigenvalue of Laplace matrix is decomposed into $O(m^3)$. Since the number of granular-balls m is far less than the original data n , the time and space efficiency of the algorithm is greatly improved.

IV. EXPERIMENT

A. Experimental Environment

Because the performance evaluation of the algorithm is involved in this paper, the experimental environment may have a certain impact on the final results. Therefore, the fairness of the experiment is fully considered in the process of the experiment. This experiment is based on the python environment.

All the experiments in this paper were carried out on a computer with 16 GB DDR4 memory and 3.0 GHz Intel (R) core (TM) i5-9500 CPU. The system environment of the computer was windows 1064 b. All the experiments were completed in pycharm 2020.2.3 (Community Edition). In order to prevent the

Algorithm 1. Generation of Granular-Balls

Input Dataset D, number of clusters k
Output GB sets

- 1 **For** each granular-ball GB_j in D **do**
- 2 calculate DM_A, DM_{weight} ,
- 3 according to (2), (3), (4),(5);
- 4 **If** $DM_{weight} \geq DM_A$ **Then**
- 5 Split GB_j ;
- 6 **End If**
- 7 **If** the number of GBs is not changing **Then**
- 8 break;
- 9 **End For**
- 10 **For** each granular-ball GB_j in D **do**
- 11 calculate $mean(r), median(r)$,
- 12 **If** $r_j \geq 2 \times \max(mean(r), median(r))$ **Then**
- 13 Split GB_j ;
- 14 **End If**
- 15 **If** the number of GBs is not changing **Then**
- 16 break;
- 17 **End For**
- 18 **return** GB sets;

reading and writing speed of the computer from affecting the experimental results, the input data of the experiment has been read and written before the test.

B. Experimental Data

In order to verify the clustering performance of GBSC algorithm in the experiment, the experiment is verified on 12

Algorithm 2. GBSC Algorithm**Input** GB sets**Output** Clustering result Y

- 1 Construct the granular-ball similarity matrix W .
- 2 Constructing non normalized Laplacian matrix L_w .
- 3 Constructing normalized Laplace matrix L_n .
- 4 Will L_n feature decomposition to calculate k minimum eigenvalues and corresponding eigenvectors.
- 5 Mapping the corresponding nodes to the k dimensional space for $k - means$ clustering to obtain the cluster partition $C_1, C_2, \dots, C_k$.

TABLE I
INFORMATION OF THE SYNTHETIC DATASETS

dataset	number of the sample	dimensionality	category
D1	944	2	2
D2	1043	2	2
D3	1039	2	4
D4	567	2	2
D5	1641	2	3
D6	876	2	2
D7	1741	2	6
D8	1016	2	4
D9	6699	2	5
D10	1427	2	4
D11	3603	2	3
D12	1020	2	3

TABLE II
INFORMATION OF THE REAL DATASETS

dataset	number of the sample	dimensionality	category
Abalone	4177	8	3
Drybean	13611	16	7
Waveform	5000	21	3
Pendigits	10992	16	10

synthetic data sets and 4 UCI [40] real data machine learning database. The details of the synthetic dataset and the real dataset are shown in Tables I and II respectively.

C. Evaluating Indicator

In order to evaluate the clustering results of different algorithms, this paper uses two evaluation indicators, NMI [41] (standard mutual information) and ARI [42] (adjusted rand index), to cluster analysis. At the same time, in order to enhance the reliability of the results, each test method will run 10 times on each data set, and then take its mean value as the final result and show it.

NMI is used to measure the similarity of two clustering results, and its calculation formula is as follows:

$$NMI = \frac{2I(Y; C)}{H(Y) + H(C)}. \quad (13)$$

$$H(X) = - \sum_{i=0}^{|X|} P(i) \log_2 P(i). \quad (14)$$

$$I(Y; C) = H(Y) - H(Y|C), \quad (15)$$

where Y represents the category of real data, C represents the result of clustering, $H(\cdot)$ represents the cross entropy. $I(Y; C)$ denotes mutual information. Mutual information is a useful information measure in information theory. It can be regarded as the amount of information about another random variable contained in a random variable, or the uncertainty of a random variable reduced by the knowledge of another random variable. In short, it represents the correlation between two time sets.

The value range of NMI is $[0, 1]$. The larger the value, the more consistent it is with the real result, that is, the better the clustering effect. ARI measures the similarity between the two clustering results by calculating the number of sample point pairs located in the same cluster and different clusters. The calculation formula is as follows:

$$ARI = \frac{2(ad - bc)}{(a + b)(b + d)(a + c)(c + d)}, \quad (16)$$

where a represents the number of point pairs that belong to the same cluster in both the real and experimental cases, b represents the number of point pairs that belong to the same cluster in the real case but do not belong to the same cluster in the experimental case, c represents the number of point pairs that do not belong to the same cluster in the real case but belong to the same cluster in the experimental case, and d represents the number of point pairs that do not belong to the same cluster in both the real and experimental cases. The value range of ARI is $[-1, 1]$, The higher the value, the better the clustering effect.

D. The Number of Granular-Balls

Using 12 synthetic data sets, the number of granular-ball in each synthetic data set is obtained as shown in Fig. 3. The similarity matrix of granular-ball spectral clustering will be composed of these granular-ball center points. The similarity matrix of traditional spectral clustering is composed of each sample point. Fig. 4 shows the comparison between the original sample points and the number of granular-ball centers. By comparing the number of granular-ball center points with the number of original sample points, it can be found that after granular-ball processing, The scale of constructing similarity matrix can be greatly reduced. For synthetic data sets, the maximum scale of the original similarity matrix can be reduced by twenty-five times, and the minimum scale of the original similarity matrix can be reduced by nine times. Fig. 5 shows the comparison between the original sampling point and the center of the granular-ball on the real data sets. It can be seen that the data size has been reduced by at least four times.

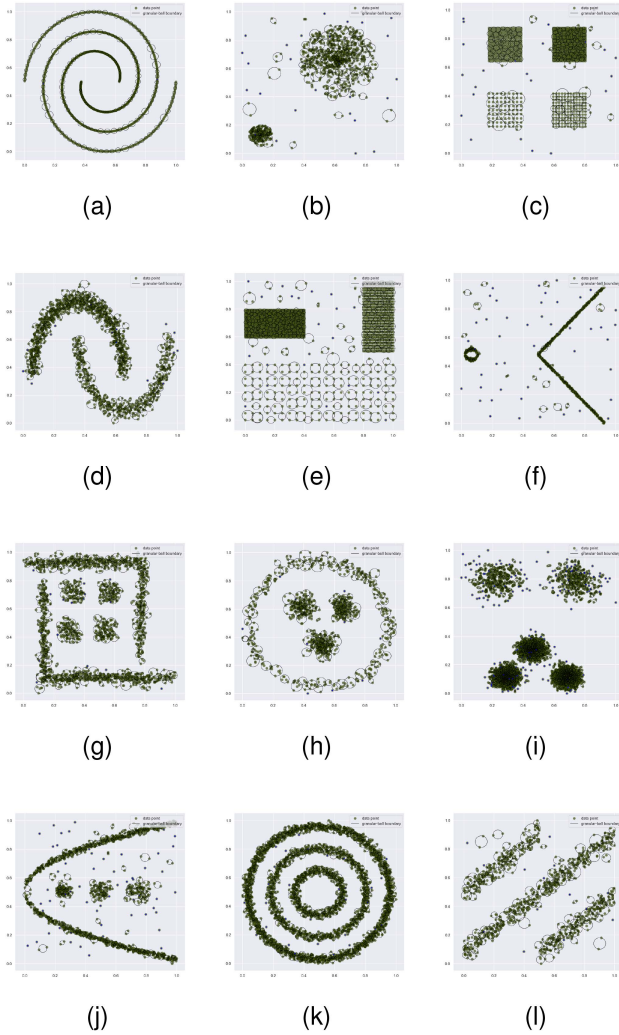


Fig. 3. The granular-ball on 12 synthetic data sets.

E. Experimental Process

In order to test the effectiveness of GBSC algorithm, this algorithm is compared with classical spectral clustering algorithm (SC), classical clustering algorithm (Kmeans) and several improved spectral clustering algorithms. The improved algorithms include USENC algorithm, USPNC [21] algorithm, LSCH [43] algorithm, DNCSC [44] algorithm, FRWL [45] algorithm, CSSP [46] algorithm and KASP [47] algorithm.

The experimental results not only give the performance comparison between this algorithm and traditional spectral clustering, but also get the performance advantages of GBSC algorithm compared with the latest clustering algorithm in large-scale data set processing.

The parameters set in this experiment are shown in Table III and Table IV.

F. Correctness and Convergence

1) *Clustering Results Comparison:* Experiments on real data sets are mainly used to evaluate the effectiveness and execution performance of the algorithm. In the experimental results, we

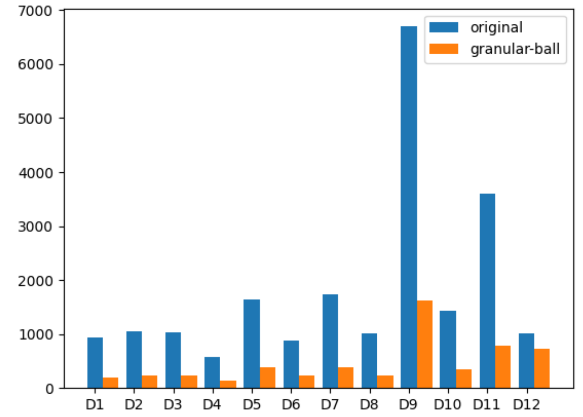


Fig. 4. Comparison between the original sampling point and the center of the granular-ball on the synthetic data sets.

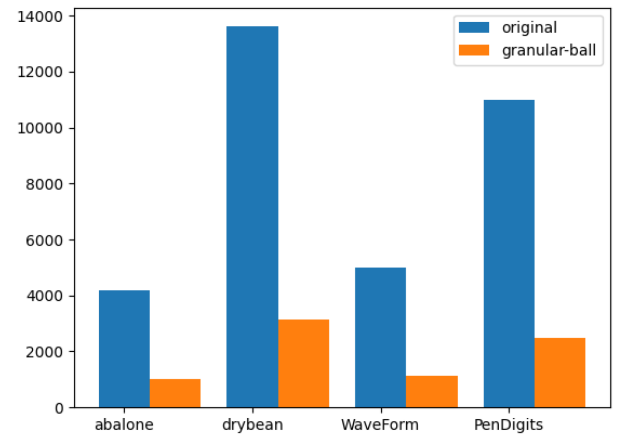


Fig. 5. The comparison between the original sampling point and the center of the granular-ball on the real data sets.

need to combine the speed and effectiveness to comprehensively evaluate the performance of the algorithm.

After using the above 10 algorithms to get the clustering results, NMI and ARI are used to analyze the above clustering results. The experimental results are shown in Table V and Table VI. On most datasets, the algorithm in this paper is better than the traditional spectral clustering algorithm in time and clustering effect, which is closer to the real partition. The clustering performance of other improved algorithms is far from that of this algorithm or even the traditional spectral clustering algorithm on some datasets, which shows that other improved algorithms are only effective for specific datasets. For some data sets, it is not as good as traditional spectral clustering, and the spectral clustering algorithm proposed in this paper is basically better than the traditional spectral clustering algorithm in clustering effect. It is a general algorithm to improve the performance of spectral clustering algorithm.

2) *Clustering Performance Comparison:* As shown in Figs. 6 and 7, when 12 synthetic datasets achieve the clustering effect similar to that of the original spectral clustering, the running time of the granular-ball spectral clustering is greatly improved compared with that of the original spectral clustering,

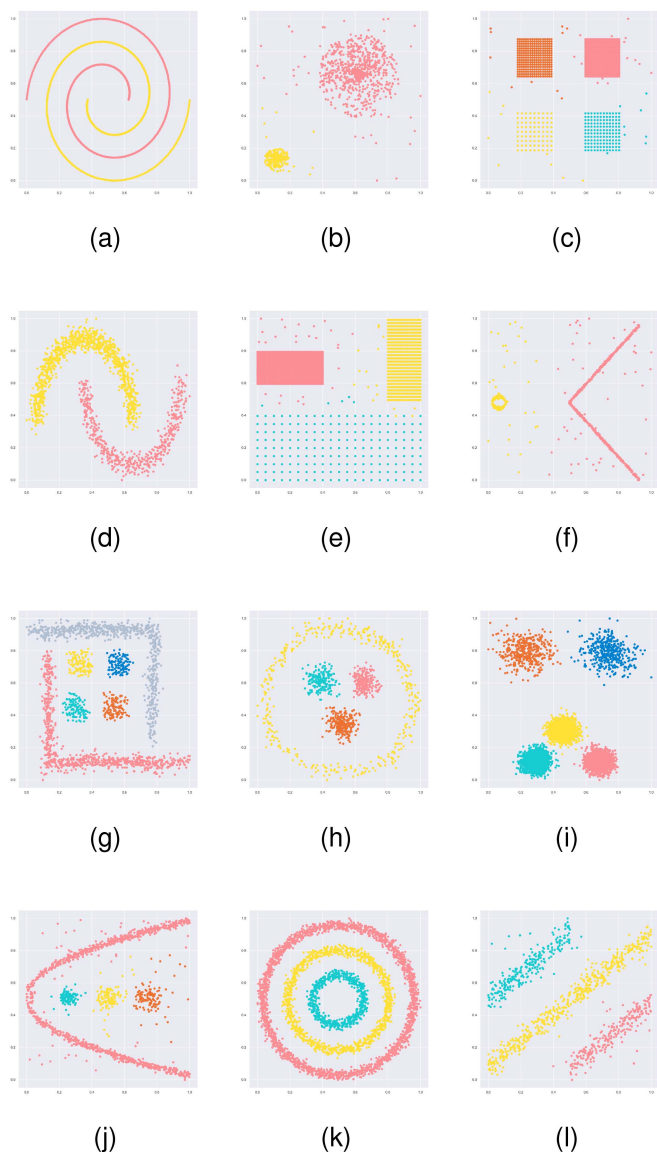


Fig. 6. The clustering results of GBSC on 12 synthetic data sets.

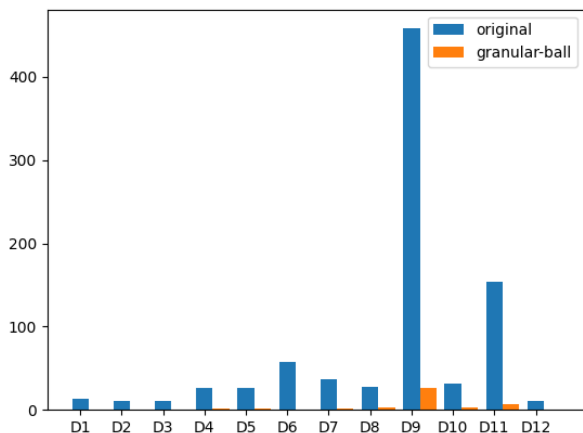


Fig. 7. Time consumption comparison.

TABLE III
PARAMETERS SETTINGS ON SYNTHETIC DATA SETS.

	USENC	USPEC	LSCH	DNCSC	FRWL
D1	k=2	k=2,m=20	k=2	k=2	k=2,c=25
D2	k=2	k=2,m=20	k=2	k=2	k=2,c=25
D3	k=4	k=4,m=20	k=4	k=4	k=4,c=25
D4	k=2	k=2,m=20	k=2	k=2	k=2,c=25
D5	k=3	k=3,m=20	k=3	k=3	k=3,c=25
D6	k=2	k=2,m=20	k=2	k=2	k=2,c=25
D7	k=6	k=6,m=20	k=6	k=6	k=6,c=25
D8	k=4	k=4,m=20	k=4	k=4	k=4,c=25
D9	k=5	k=5,m=20	k=5	k=5	k=5,c=25
D10	k=4	k=4,m=20	k=4	k=4	k=4,c=25
D11	k=3	k=3,m=20	k=3	k=3	k=3,c=25
D12	k=3	k=3,m=20	k=3	k=3	k=3,c=25

	Kmeans	CSSP	KASP	SC	GBSC
D1	k=2	k=2,m=500	k=2,gama=2	k=2,deta=0.1	k=2,deta=0.05
D2	k=2	k=2,m=500	k=2,gama=2	k=2,deta=0.1	k=2,deta=0.1
D3	k=4	k=4,m=500	k=4,gama=2	k=4,deta=0.1	k=4,deta=0.1
D4	k=2	k=2,m=500	k=2,gama=2	k=2,deta=0.1	k=2,deta=0.1
D5	k=3	k=3,m=500	k=3,gama=2	k=3,deta=0.1	k=3,deta=0.1
D6	k=2	k=2,m=500	k=2,gama=2	k=2,deta=0.1	k=2,deta=0.1
D7	k=6	k=6,m=500	k=6,gama=2	k=6,deta=0.1	k=6,deta=0.1
D8	k=4	k=4,m=500	k=4,gama=2	k=4,deta=0.1	k=4,deta=0.05
D9	k=5	k=5,m=500	k=5,gama=2	k=5,deta=0.1	k=5,deta=0.1
D10	k=4	k=4,m=500	k=4,gama=2	k=4,deta=0.1	k=4,deta=0.05
D11	k=3	k=3,m=500	k=3,gama=2	k=3,deta=0.1	k=3,deta=0.05
D12	k=3	k=3,m=500	k=3,gama=2	k=3,deta=0.1	k=3,deta=0.1

TABLE IV
PARAMETERS SETTINGS ON UCI DATA SETS.

	USENC	USPEC	LSCH	DNCSC	FRWL
Abalone	k=3	k=3,m=20	k=3	k=3	k=3,c=25
Drybean	k=7	k=7,m=20	k=7	k=7	k=7,c=25
Waveform	k=3	k=3,m=20	k=3	k=3	k=3,c=25
Pendigits	k=10	k=10,m=20	k=10	k=10	k=10,c=25

	Kmeans	CSSP	KASP	SC	GBSC
Abalone	k=3	k=3,m=500	k=3,gama=2	k=3,deta=0.1	k=3,deta=0.1
Drybean	k=7	k=7,m=500	k=7,gama=2	k=7,deta=0.1	k=7,deta=0.1
Waveform	k=3	k=3,m=500	k=3,gama=2	k=3,deta=0.1	k=3,deta=0.1
Pendigits	k=10	k=10,m=500	k=10,gama=2	k=10,deta=0.1	k=10,deta=0.1

with a minimum reduction of 9 times and a maximum reduction of 104 times. The results of the other comparison algorithms on the 12 artificial datasets are shown in Figs. 1–9 in the supplementary materials, (*available online*). The reduction of running time is related to the data set size and data distribution. The more distinct the separation between clusters, the more concentrated the density center, and the more significant the reduction.

As shown in Fig. 8, when 12 synthetic datasets achieve the clustering effect similar to that of the original spectral clustering, the storage space consumed by the granular-ball spectral clustering is greatly reduced compared with that of

TABLE V
COMPARISON OF NMI VALUES OF DIFFERENT ALGORITHMS

dataset	USENC	USPEC	LSCH	DNCSC	FRWL	KMEANS	CSSP	KASP	SC	GBSC
Abalone	0.042	0.010	0.069	0.012	0.136	0.102	0.132	0.121	0.165	0.166
Drybean	0.508	0.161	0.467	0.160	0.424	0.387	0.508	0.496	0.711	0.730
Waveform	0.367	0.371	0.370	0.368	0.369	0.368	0.368	0.365	0.368	0.368
Pendigits	0.858	0.802	0.822	0.792	0.687	0.587	0.478	0.558	0.597	0.597

TABLE VI
COMPARISON OF ARI VALUES OF DIFFERENT ALGORITHMS

dataset	USENC	USPEC	LSCH	DNCSC	FRWL	KMEANS	CSSP	KASP	SC	GBSC
Abalone	0.011	0.012	0.074	0.005	0.125	0.073	0.136	0.080	0.124	0.164
Drybean	0.338	0.032	0.302	0.032	0.282	0.238	0.367	0.282	0.643	0.669
Waveform	0.250	0.256	0.259	0.254	0.253	0.250	0.250	0.257	0.250	0.251
Pendigits	0.790	0.717	0.753	0.650	0.602	0.436	0.366	0.402	0.424	0.426

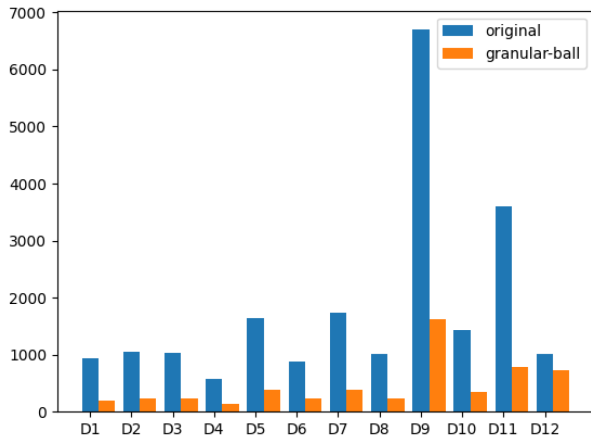


Fig. 8. Memory consumption comparison.

the original spectral clustering, at least twice as much as the storage space, and at most 13 times as much as the storage space. The reduction of storage space is related to the size of the dataset. The larger the dataset, the more obvious the reduction of storage space. It can be seen that granular-ball spectral clustering can not only improve the clustering effect, but also greatly improve the clustering performance of traditional spectral clustering.

V. CONCLUSION

Aiming at the high computational cost of spectral clustering algorithm, this paper proposes an improved spectral clustering algorithm based on granular-ball. This algorithm is based on the idea of multi granularity. It describes the unsupervised data using granular-ball, and then constructs the similarity matrix of spectral clustering based on granular-balls. Thus, the scale of matrix calculation is greatly reduced, and the time and space

efficiency of spectral clustering method are improved. Experiments show that the algorithm achieves good results in both time and accuracy.

Although this algorithm has made some progress in reducing the time complexity and spatial complexity of spectral clustering and improving the effect of spectral clustering. There are several improvement directions for the method proposed in this paper. This paper only optimizes the construction of the similarity matrix, and has no further research on how to optimize the solution of the similarity matrix. At the same time, parallel computing can also be introduced to further accelerate the computational efficiency of the spectral clustering algorithm. In the future work, the granular-ball based spectral clustering algorithm can be combined with other improved spectral clustering algorithms, so that the spectral clustering algorithm can achieve better performance on large-scale data sets.

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